TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	57.2 ± 5.31
Correlation Stability	80.5 ± 0.81
Distribution Consistency	68.1 ± 13.92
Entropy Preservation	-74.3 ± 6.65
Mean Z Score Drift	0.5 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	1.7 ± 0.90
Data Density	100.0 ± 0.00
Coverage Score	93.9 ± 1.51
Feature Accuracy	65.9 ± 0.70
Novelty Score	0.7 ± 0.02

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Dataset Audit: hepatitis.csv

1. Dataset Portrait

Attribute	Value
Rows	155
Columns	20
Numeric features	20
Categorical features	0
Missing cells (%)	5.4%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	41.200	45.658	+4.458	10.8%	Warn
Sex	1.103	1.000	-0.103	9.4%	OK
Steroid	1.506	1.367	-0.139	9.2%	OK
Antivirals	1.845	1.959	+0.114	6.2%	OK
Fatigue	1.351	1.043	-0.307	22.8%	Warn
Malaise	1.604	1.272	-0.332	20.7%	Warn
Anorexia	1.792	1.594	-0.198	11.1%	Warn
Liver Big	1.828	1.911	+0.084	4.6%	OK
Liver Firm	1.583	1.724	+0.141	8.9%	OK
Spleen Palpable	1.800	1.573	-0.227	12.6%	Warn
Spiders	1.660	1.283	-0.377	22.7%	Warn
Ascites	1.867	1.487	-0.380	20.3%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	57.2 ± 5.31
Correlation Stability (*2)	80.5 ± 0.81
Distribution Consistency (*3)	68.1 ± 13.92
Entropy Preservation (*4)	-74.3 ± 6.65
Mean Z Score Drift (*5)	0.5 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	1.7 ± 0.90

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	93.9 ± 1.51
Feature Accuracy (*10)	65.9 ± 0.70
Novelty Score (*11)	0.7 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8064516129032258	N/A	0.5419354838709677	-0.2645161290322581	*
F1	N/A	N/A	0.3587487950645845 7	0.35874879506458457	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	N/A	0.832258064516129	-0.00645161290322579	
F1	0.5718392134181608	N/A	0.5807782276203328	0.008939014202172102	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8548387096774194	N/A	0.8354838709677418	-0.01935483870967744	
F1	0.5367132867132867	N/A	0.5753717850776673	0.03865849836438076	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	N/A	0.5161290322580645	-0.2645161290322581	*
F1	N/A	N/A	0.2859035548738066	0.2859035548738066	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8096774193548386	N/A	0.8096774193548386	2.2204460492503132e-17	
F1	0.3897757213546687	N/A	0.5772145632145632	0.18743884185989448	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7838709677419355	N/A	0.7709677419354839	-0.012903225806451613	
F1	0.4513442113442114	N/A	0.4733065953654189	0.021962384021207555	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8548387096774193	N/A	0.8290322580645161	-0.025806451612903236	
F1	0.5919717963196224	N/A	0.5432539682539683	-0.04871782806565415	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	N/A	0.7741935483870968	-0.016129032258064512	
F1	0.561449871078354	N/A	0.5519739806117514	-0.009475890466602549	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	N/A	0.7903225806451613	-0.04838709677419356	
F1	0.5132478632478632	N/A	0.535958485958486	0.022710622710622722	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	N/A	0.8483870967741934	-0.0032258064516129227	
F1	0.5385593354014405	N/A	0.5925751441540916	0.05401580875265084	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
2	50.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
8	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
9	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
10	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
11	39.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0
12	32.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
13	41.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
14	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
15	47.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
16	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
17	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
18	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
19	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
20	38.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	49.0	1.0	2.0	2.0	1.0	1.3727369809671341	1.3727369809671341	2.0	2.0	2.0
2	53.0	1.0	1.4347743362248178	2.0	1.0	1.5652256637751822	1.0	2.0	2.0	1.4347743362248178
3	57.0	1.0	1.9492798546690855	2.0	1.0	1.0	1.0507201453309145	2.0	2.0	2.0
4	43.0	1.0	1.4365394987517268	2.0	1.5634605012482732	2.0	2.0	2.0	1.0	1.5634605012482732
5	31.0	1.0	1.7758651719209542	2.0	1.2241348280790458	1.2241348280790458	1.2241348280790458	2.0	1.0	2.0
6	38.0	1.0	1.0	2.0	1.9996383745965773	1.9996383745965773	2.0	2.0	1.0003616254034227	1.9996383745965773
7	61.0	1.0	1.0	2.0	1.0	1.0	2.0	nan	nan	2.0
8	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
9	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
10	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	41.0	1.0	2.0	2.0	1.0	2.0	2.0	1.9842840333065113	1.0	1.0
2	54.0	1.0	2.0	2.0	1.0	1.7284398462880515	1.7284398462880515	1.2715601537119485	1.2715601537119485	1.2715601537119485
3	49.0	1.0	1.1028481133756287	2.0	1.0	1.1028481133756287	2.0	1.8971518866243713	1.0	1.8971518866243713
4	48.0	1.0	1.8357653544759152	2.0	1.0	1.0	2.0	2.0	1.1642346455240848	1.8357653544759152
5	47.0	1.0	1.8676417052614696	2.0	1.0	1.0	2.0	2.0	1.1323582947385304	1.8676417052614696
6	47.0	1.0	2.0	2.0	1.3173534026693194	1.3173534026693194	2.0	2.0	1.3173534026693194	2.0
7	60.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
9	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	49.0	1.0	2.0	2.0	1.0	1.7991511859613423	2.0	2.0	1.0	1.2008488140386577
2	48.0	1.0	2.0	2.0	1.0	1.3490547091455856	2.0	2.0	1.0	1.6509452908544144
3	49.0	1.0	1.5018644565357705	2.0	1.0	1.5018644565357705	2.0	2.0	1.4981355434642205	1.0
4	46.0	1.0	1.4094518932604878	2.0	1.0	1.4094518932604878	2.0	2.0	1.5905481067395122	1.0
5	46.0	1.0	1.8917497493692732	1.0	1.0	1.0	2.0	2.0	1.0	2.0
6	35.0	1.0	1.0	1.0	1.0	1.7891162157415987	2.0	2.0	1.7891162157415987	1.2108837842584013
7	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	31.0	1.0	1.0058747200502394	2.0	1.0	1.0	1.0	2.0	2.0	1.0058747200502394
2	31.0	1.0	1.0	2.0	1.0	1.0	1.1483228951433895	2.0	2.0	1.1483228951433895
3	43.0	1.0	1.0	2.0	1.0	1.0	1.6983438425398525	2.0	2.0	1.0
4	35.0	1.0	1.0	2.0	1.0	1.8523900952531203	1.8523900952531203	2.0	1.8523900952531203	1.1476099047468797
5	51.0	1.0	1.0	2.0	1.0	1.1292154637933405	2.0	2.0	2.0	1.0
6	36.0	1.0	1.0	2.0	1.0	1.4860137831250835	1.4860137831250835	2.0	1.4860137831250835	1.5139862168749165
7	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
9	25.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
10	38.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	45.0	1.0	2.0	2.0	1.0	1.47188833414579 5	2.0	2.0	1.47188833414579 5	1.52811166585420 5
2	43.0	1.0	2.0	2.0	1.0	1.96882692302478 04	2.0	2.0	1.96882692302478 04	1.03117307697521 96
3	46.0	1.0	1.85443255046656 09	2.0	1.0	1.0	1.85443255046656 09	2.0	1.14556744953343 91	1.85443255046656 09
4	57.0	1.0	1.0	2.0	1.0	1.0	1.51425472176551	1.51425472176551	1.0	1.48574527823449
5	48.0	1.0	1.52765182998956 78	2.0	1.0	1.52765182998956 78	1.52765182998956 78	1.52765182998956 78	1.0	1.47234817001043 22
6	56.0	1.0	1.0	2.0	1.0	1.0	1.30524251607861 34	1.30524251607861 34	1.0	1.69475748392138 66
7	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	33.0	1.0	1.25434938275738 88	2.0	1.0	1.74565061724261 12	1.74565061724261 12	2.0	1.74565061724261 12	1.25434938275738 88
2	37.0	1.0	2.0	2.0	1.0	1.0	1.42400025861552 98	2.0	1.0	2.0
3	37.0	1.0	1.0	2.0	1.0	1.0	1.85110528028193 6	1.0	1.0	2.0
4	40.0	1.0	1.87165200257774 97	2.0	1.0	1.87165200257774 97	2.0	1.87165200257774 97	1.0	1.12834799742225 03
5	54.0	1.0	1.85386958824593 9	2.0	1.0	1.85386958824593 9	2.0	1.0	1.0	1.14613041175406 1
6	41.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0
7	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
8	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
9	51.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
10	40.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	46.0	1.0	2.0	2.0	1.0	1.090784705500372	2.0	2.0	1.090784705500372	2.0
2	45.0	1.0	2.0	2.0	1.0	1.255813305969208	2.0	2.0	1.255813305969208	1.744186694030792
3	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	1.617176726789259
4	41.0	1.0	1.1687732653542333	2.0	1.0	1.1687732653542333	1.1687732653542333	2.0	2.0	1.1687732653542333
5	37.0	1.0	1.9898324337813893	2.0	1.0	1.9898324337813893	2.0	2.0	2.0	2.0
6	53.0	1.0	1.3148944413523886	2.0	1.0	1.3148944413523886	2.0	2.0	2.0	2.0
7	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
9	30.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	56.0	1.0	1.0	2.0	1.0	1.0	1.2949295455270282	1.2949295455270282	1.0	1.7050704544729718
2	57.0	1.0	1.0	2.0	1.0	1.0	1.6167314147225142	1.6167314147225142	1.0	1.3832685852774858
3	50.0	1.0	1.5321537710187156	2.0	1.0	1.0	2.0	2.0	1.4678462289812844	1.5321537710187156
4	47.0	1.0	2.0	2.0	1.2553389143445632	1.2553389143445632	2.0	2.0	1.2553389143445632	2.0
5	40.0	1.0	1.1610041112958252	2.0	1.0	1.0	1.0	2.0	1.8389958887041748	1.1610041112958252
6	33.0	1.0	2.0	2.0	1.0	1.0	1.1786192023596784	2.0	1.0	2.0
7	65.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
8	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
9	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
10	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	49.0	1.0	1.9923402819160743	2.0	1.0	1.9923402819160743	2.0	2.0	1.0076597180839257	1.0
2	37.0	1.0	1.0	2.0	1.0	1.6872139504120653	2.0	2.0	2.0	1.0
3	54.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0
4	55.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.758440740193469
5	53.0	1.0	1.0168839855081397	2.0	1.0168839855081397	1.0168839855081397	2.0	2.0	2.0	1.0168839855081397
6	43.0	1.0	1.0	2.0	1.0	1.0	1.3340350801968455	2.0	1.3340350801968455	1.6659649198031545
7	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	44.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
9	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	55.0	1.0	1.1807623210665068	2.0	1.0	1.0	2.0	2.0	1.8192376789334932	2.0
2	47.0	1.0	1.9244965314540023	2.0	1.0	1.0	2.0	2.0	1.0755034685459977	1.9244965314540023
3	53.0	1.0	1.3405679140259168	2.0	1.0	1.0	2.0	2.0	1.6594320859740832	2.0
4	45.0	1.0	2.0	2.0	1.0	1.4480695664695068	2.0	2.0	1.4480695664695068	1.551930433530492
5	42.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0
6	57.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0
7	56.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	28.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	49.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0